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Exploring the role of renewable energy communities in the Italian flexibility markets

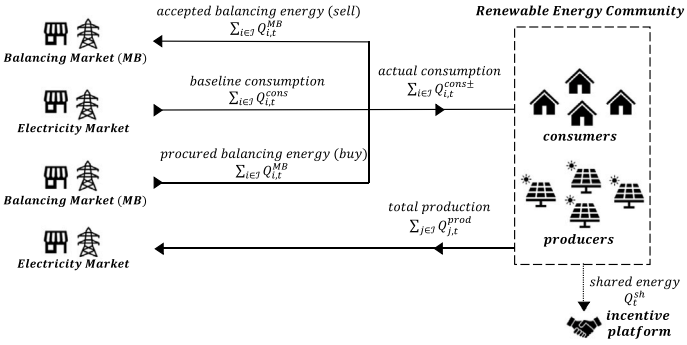
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HIGHLIGHTS

- A simulation framework is developed to model Renewable Energy Communities participation in the Ancillary Services Market.
- Demand-side flexibility from Renewable Energy Communities is analysed using real data.
- Multiple flexibility levels are evaluated in terms of balancing revenues and shared energy.
- Flexibility revenues largely compensate reductions in shared energy incentives.
- Economically attractive results are achieved even at moderate flexibility levels.

GRAPHICAL ABSTRACT



Description: We propose a simulation framework for flexible load management in a Renewable Energy Community (REC) participating in the balancing market. By shifting consumption upward and downward according to market prices and flexibility levels, we demonstrate significant ancillary service revenues while maintaining high shared energy within the community. The diagram summarizes load allocation, market interaction, and resulting economic benefits.

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ABSTRACT

The increasing penetration of distributed generation, mainly from variable and scarcely predictable renewable energy sources, requires new forms of system flexibility. Renewable Energy Communities (RECs) can contribute by aggregating demand-side flexibility and participating in ancillary service markets. This paper analyses the participation of an Italian REC in the Ancillary Services Market, focusing on load flexibility and its interaction with the shared energy incentive scheme. Using real consumption and generation data, multiple flexibility scenarios are simulated and evaluated in terms of balancing revenues, shared energy, and overall community cash flows. Results show that flexibility provision is economically attractive even at moderate levels, while the impact on shared energy remains limited and, in some cases, positive. The study confirms the potential of RECs as effective flexibility providers in decentralised electricity systems.

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[1] This paper represents the view of the authors. It is not intended to represent the position or opinions of any company with which the authors have an employment relationship.

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Glossary of Acronyms

aFRR Automatic Frequency Restoration Reserve
ARERA Italian Regulatory Authority for Energy, Networks and Environment
ASM Ancillary Services Market
BESS Battery Energy Storage System
BRP Balancing Responsible Party
BSP Balancing Service Provider
CF Cash Flow
CO_2 Carbon Dioxide
DER Distributed Energy Resources
DG Distributed Generation
DNO Distribution Network Operator
DSO Distribution System Operator
EC Energy Community
EM Electricity Market
ex-ante MSD Ex-ante Dispatching Services Market (*Mercato dei Servizi di Dispacciamento ex ante*)
FCR Frequency Containment Reserve
GME Gestore dei Mercati Energetici
GSE Gestore dei Servizi Energetici
MB Balancing Market (*Mercato del Bilanciamento*)
MBR Balancing and Redispatching Market (*Mercato del Bilanciamento e del Ridispacciamento*)
mFRR Manual Frequency Restoration Reserve
MSD Dispatching Services Market (*Mercato dei Servizi di Dispacciamento*)
PV Photovoltaic
PUN National Single Electricity Price (*Prezzo Unico Nazionale*)
REC Renewable Energy Community
RID Dedicated Withdrawal Scheme (*Ritiro Dedicato*)
RR Replacement Reserve
TIDE Integrated Dispatching Text (*Testo Integrato del Dispacciamento Elettrico*)
TIP Incentive Tariff for Shared Energy
TRAS Transmission Tariff
TSO Transmission System Operator
UVA Enabled Virtual Unit (*Unità Virtuale Abilitata*)
UVAM Mixed Enabled Virtual Unit (*Unità Virtuale Abilitata Mista*)

1. Introduction

1.1. Contextual overview

The European energy system is undergoing a rapid transformation driven by CO_2 reduction targets established in Regulation (EU) 2021/1119 [1], known as the European Climate Law, a cornerstone of the European Green Deal. To support these ambitions, the Fit for 55 package updates climate and energy legislation [2]. This evolving framework reshapes generation, markets, transmission, distribution, and consumer engagement.

The widespread development of renewable sources, especially non-programmable ones, the ever-increasing diffusion of Distributed Generation (DG) and the expected increase in final demand, caused by the strong electrification of consumption, are leading to a complete revision of the system management paradigm, which inevitably has repercussions on the functioning of electricity markets and ancillary services [3,4].

Traditional energy models are no longer adequate, prompting urgent development of new solutions to enhance system flexibility. Advances in technology are transforming energy production, consumption, and balancing mechanisms via the development of smart grids and new ways of procuring ancillary services [3,5].

Flexibility has traditionally been a key component of power system operation. However, the increasing penetration of renewable energy sources and the progressive reduction of conventional dispatchable generation are driving the need for new flexibility sources, including demand-side flexibility, to address growing challenges in transmission and distribution system management, such as network balancing, line congestion, and voltage regulation [6].

A user is defined as flexible if, at a specific operating point, it is able to respond to a series of uncertain future events (such as requests for modulation of energy supplied/withdrawn), taking relevant actions within an acceptable cost threshold and time frame [7,8]. The Clean Energy Package [3], indicates the adoption of market-based solutions to purchase flexibility resources to meet these needs as an alternative solution to network reinforcement.

In parallel, the directive [9] promotes the active participation of consumers—individually or collectively via energy communities—in all energy markets.

In Italy, with the new integrated electricity dispatching framework, promoted by Integrated Dispatching Text (TIDE) (Testo Integrato del Dispacciamento Elettrico) [4] a series of significant changes has been introduced aimed at enhancing market efficiency, flexibility, and participation. Among these, global ancillary services have been opened both to consumption units and to production units with a capacity of less than 10 MW, marking a substantial broadening of market access, enabling their participation through aggregation mechanisms. Such aggregations must satisfy the technical requirements defined by the regulation, including a minimum modulation capacity of 1 MW. Additionally, the framework allows for the establishment of Enabled Virtual Units (UVAs), facilitating aggregated participation in the electricity system. Furthermore Distribution System Operator (DSO) [2] resources can now also be activated through local flexibility markets, promoting a more dynamic and decentralised grid management.

This study presents a simulation-based approach that combines user-level modelling of flexible loads in RECs with an integrated economic assessment of ancillary service revenues and shared energy incentives, the latter specifically considering the Italian regulatory framework. Rather than introducing a novel mathematical formulation, the proposed framework provides a detailed representation of load shifting and its economic consequences for the REC under different flexibility-availability scenarios. In particular, it enables the interaction between balancing-market participation and internal energy sharing to be assessed jointly, highlighting the potential for RECs to provide flexibility through aggregation mechanisms while retaining a substantial share of their energy-sharing benefits.

1.2. Literature analysis

The growing interest in Renewable Energy Communities (RECs) has generated an extensive body of literature addressing their design, operation, market participation, and provision of flexibility services. However, existing studies investigate these topics from different perspectives and often focus on specific aspects of REC operation, making it difficult to identify the remaining research challenges.

To provide a structured overview, the literature is organised into three complementary themes. First, recent studies investigating the role

[2] In this scenario, the Distribution Network Operator (DNO) is called upon to innovate its role, expanding its scope beyond the management of network infrastructures to become a buyer of network services from resources connected to its network, thus transforming itself into a real DSO.

of RECs as providers of flexibility services are reviewed. Second, the Italian regulatory framework governing ancillary service participation is introduced, since it represents the operational context considered in this work. Finally, the main research gaps emerging from the literature are discussed to position the contribution of the present study.

1.2.1. RECs for unlocking decentralised system flexibility

Renewable Energy Communities (RECs), introduced by Directive (EU) 2018/2001 and complemented by Directive (EU) 2019/944, are legal entities that enable citizens, small and medium-sized enterprises, and local authorities to collectively generate, consume, store, share, and manage renewable electricity. Rather than pursuing profit maximisation, RECs are established to deliver environmental, economic, and social benefits to their members and the local area. Operationally, RECs aggregate distributed energy resources and coordinate the interaction between local generation and electricity demand. Community members may participate as consumers, producers, or prosumers, while the REC enables energy sharing and coordinated resource management. Subject to the applicable regulatory framework, RECs may also participate in electricity and flexibility markets through aggregation, demand response, and ancillary service provision, thereby supporting the transition towards increasingly decentralised and flexible power systems.

In the Italian regulatory framework, RECs can benefit from an incentive mechanism based on the so-called shared energy. This term refers to the amount of renewable electricity produced by REC members and virtually consumed by other members of the community during the same time interval (1 h) [10]. The mechanism does not involve a physical exchange of electricity among members, but rather a virtual matching between renewable generation and local consumption. The shared energy determines the eligibility for an incentive tariff (TIP), which represents one of the main economic benefits associated with REC participation.

The growing diffusion of RECs has stimulated extensive research on their planning and operation. Existing studies have investigated different community configurations and business models, considering economic, environmental, and social objectives as well as the trade-offs arising from coordinated operation [11]. In parallel, increasing renewable penetration and distribution network constraints have progressively shifted the research focus from energy sharing alone towards the exploitation of flexibility as an additional service that RECs can provide to the electrical system.

A first group of studies investigates flexibility as a mechanism to improve the internal operation of energy communities. Several works optimise flexible demand, distributed generation, and storage to maximise self-consumption, increase shared energy, and improve the overall economic performance of the community. In the Italian context, end-user flexibility has been exploited to increase the value of shared energy through demand scheduling algorithms and techno-economic analyses, demonstrating the benefits of coordinating consumption with renewable generation profiles [12]. Similarly, demand-side flexibility has been investigated to increase the economic value of energy sharing under the Italian regulatory framework through load-shifting strategies and optimisation-based approaches [13]. Other contributions have explored the role of flexibility in improving users' wellbeing and community resilience by coordinating controllable household loads within microgrid-based energy communities [14]. These studies demonstrate the benefits of coordinated flexibility for community operation; however, flexibility is generally considered as an internal resource aimed at improving self-consumption, shared energy, or resilience, rather than as an external market product.

A second research direction considers RECs as providers of flexibility services for electricity networks. In this context, flexibility is exploited not only to optimise community self-consumption but also to support transmission and distribution system operation through congestion management, voltage regulation, and peak-load reduction. Recent studies have analysed how different coordination mechanisms between Distribution System Operators (DSOs) and energy communities can reduce grid congestion and defer network reinforcement while improving overall system efficiency [15]. Other works have investigated the provision of ancillary services, such as reactive power support, through coordinated operation of distributed energy resources, batteries, and local energy markets under uncertainty [16]. Similarly, optimisation strategies integrating renewable generation, storage, controllable loads, and sector-coupling technologies have shown that RECs can provide flexibility services while enabling additional revenue streams [17]. These studies highlight the increasing role of RECs as distributed flexibility providers capable of supporting secure grid operation.

More recently, several authors have investigated the economic opportunities arising from flexibility markets. Local flexibility markets enable distributed energy resources to deliver cost-effective services to system operators while creating additional revenue streams for community members. Studies have analysed the participation of energy communities in decentralised flexibility mechanisms, highlighting their potential to increase economic benefits and foster active user participation in electricity markets [18,19]. Consequently, planning methodologies have started to incorporate flexibility provision already during the sizing of distributed energy resources, considering batteries, electric vehicles, thermal assets, and expected flexibility revenues as investment drivers. For example, [20] proposes a two-stage planning methodology for sizing REC resources while considering collective self-consumption and participation in local flexibility markets, including cross-sector flexible assets such as electric vehicles and thermal loads. Similarly, studies have analysed the financial viability of energy communities participating in balancing markets, showing that flexibility revenues may significantly affect the profitability of investments in renewable generation and storage systems [21,22].

Beyond the technical and economic aspects, recent literature has also investigated the organisational and market mechanisms required to enable large-scale flexibility participation. Aggregation frameworks and coordination platforms have been proposed to connect consumers, flexibility providers, and system operators, supporting the development of flexibility-centric value chains and enabling energy communities to participate in local flexibility markets [23]. Other works have investigated different flexibility activation mechanisms, including implicit coordination among community members and explicit market-based activation by DSOs, highlighting the need to understand the interaction between these approaches in future distribution systems [24].

Furthermore, existing contributions on REC flexibility have mainly focused on supply-side resources, particularly photovoltaic generation and energy storage systems, which are often considered the main enablers of flexible operation [25]. Conversely, demand-side flexibility based on load shifting remains comparatively less explored, especially when considered as a resource for participation in ancillary service markets. Studies addressing load scheduling and demand flexibility generally focus on improving self-consumption or the economic performance of the REC rather than analysing the simultaneous impact on energy-sharing incentives and flexibility revenues [12,13]. In addition, several contributions rely on aggregated models, simplified assumptions, or synthetic datasets, limiting the assessment of flexibility provision under realistic operational and regulatory conditions.

Overall, the literature demonstrates that RECs are progressively evolving from passive self-consumption schemes towards active flexibility providers capable of interacting with electricity markets and supporting power system operation. Nevertheless, existing studies mainly analyse either the optimisation of energy sharing within the community or the provision of flexibility services as an independent objective. In particular, studies focusing on demand-side flexibility through load shifting generally aim at increasing shared energy and self-consumption, while works addressing ancillary and flexibility markets often focus on additional revenues from external services without explicitly considering the impact on community energy-sharing incentives. Therefore, the interaction between these two revenue streams remains insufficiently

investigated. In particular, an analysis of how load-shifting participation in ancillary service markets affects REC cash flows, shared energy levels, and the allocation of flexibility among community members is still missing. This work addresses this gap by analysing the impact of REC participation in ancillary service markets based on demand-side load shifting, while explicitly quantifying the resulting effects on energy sharing and community economic performance.

1.2.2. Italian ancillary services market

The ongoing transformation of the Italian electricity system has progressively extended the provision of ancillary services beyond conventional power plants, enabling distributed energy resources to actively support system balancing [4,26]. Within this framework, the Italian Ancillary Services Market (ASM) represents the main mechanism through which flexibility is procured to guarantee the secure and reliable operation of the power system [4].

In the Italian wholesale electricity market, the ASM is organised around the Mercato del Bilanciamento e del Ridispacciamento (MBR), in which Terna, the national Transmission System Operator (TSO), procures global ancillary services essential for the management of the entire electricity system, ensuring standards of safety, adequacy, quality, and efficiency [4,27]. This market is divided into a planning phase (ex-ante MSD) and an implementation phase (Balancing Market, MB) [4,27]. In ex-ante MSD, Terna procures the resources necessary for the management and control of the system and resolves any intrazonal congestion and, in the MB, activates them in real time to guarantee the balance between injections and withdrawals ("balancing").

The key ancillary services requested by the TSO include the Frequency Containment Reserve (FCR), the automatic Frequency Restoration Reserve (aFRR), the manual Frequency Restoration Reserve (mFRR) and the Replacement Reserve (RR) [4]. Unlike system marginal price energy markets, the MB operates under a pay-as-bid system in which the TSO remunerates the activated services at their offer price [27]. In this context, upward offers (sell bids) are submitted by generation or load units that are ready to increase their production and/or reduce their consumption, in response to a grid shortfall; accepting these offers represents a cost for the TSO. Instead, downward offers (buy bids) involve units available to reduce generation or increase consumption when there is excess supply on the grid, thus generating revenue for the TSO. In both cases, these ancillary services are voluntarily offered by dispatchable resources to support real-time grid balancing [27] and to generate net revenues for them.

In Italy, this framework was innovated, in an experimental form, in 2017 with the launch of the pilot project Unità Virtuali Abilitate Miste (UVAM) [26], enabling heterogeneous aggregations of Distributed Energy Resources (DERs)—including both generation and consumption units—to participate in the MB [28,29]. The initiative marked the first opening of the ASM to distributed resources, introducing a capacity remuneration mechanism, as an incentive to participate, in addition to service payments, and lowering the minimum participation threshold to 1 MW, thereby granting access to smaller units, renewable sources, and aggregators. UVAMs can provide multiple ASM services, including balancing; with their introduction a clear distinction was established between the Balancing Responsible Party (BRP), responsible for unit injections, withdrawals, and imbalance settlement between expected and actual power profiles, and the Balancing Service Provider (BSP), responsible for offering and delivering flexibility services. This experience has guided the new regulation, promoted by TIDE [4], which opens participation in MBR to aggregates (UVA) of consumption and production units, with an overall modulating power of 1 MW.

1.2.3. Research gap and contributions

As discussed in the previous sections, existing literature has extensively investigated the role of RECs as both energy-sharing entities and potential providers of flexibility services. However, these two aspects are generally analysed separately. Studies focusing on energy sharing mainly aim at maximising self-consumption, shared energy, or economic benefits derived from incentive schemes, while contributions addressing flexibility markets primarily evaluate the technical feasibility and economic profitability of providing services to system operators. Consequently, the interaction between internal energy-sharing mechanisms and the external flexibility market participation remains insufficiently explored.

In particular, the impact of flexibility activation on the cash flows in REC members, the variation of shared energy levels, and the allocation of flexibility requests among community members has not been fully quantified. This aspect is particularly relevant for demand-side flexibility, where the activation of flexible loads may improve the revenues associated with ancillary services but may at the same time reduce the simultaneity between renewable generation and local consumption, affecting the benefits associated with energy sharing.

To address this research gap, this work analyses the effects of REC participation in ancillary service markets, explicitly considering the trade-off between internal energy sharing and external flexibility provision. The proposed framework evaluates the impact of flexibility activation through the following indicators:

1. **Cash flows:** the emerging REC's savings, earnings or costs, on the energy and flexibility markets, including the reward for energy sharing
2. **Shared energy:** the percentage of REC's generation that is shared within the community
3. **Shift distribution:** the frequency of provision of upward and downward flexibility (per user)
4. **Flexibility amount:** the yearly energy quantities sold or bought by the REC on the ASM

The contribution of this work does not rely on proposing a new demand response scheduling algorithm. Instead, the novelty lies in integrating user-level demand-side flexibility modelling with the economic assessment of Renewable Energy Community participation in ancillary service markets. This integration enables the quantification of the trade-off between internal REC benefits associated with energy sharing and external revenues from flexibility provision.

The main contributions of this work can therefore be summarised as follows:

- a modelling framework for the simultaneous evaluation of REC energy sharing and participation in ancillary service markets;
- a quantitative analysis of the trade-off between internal community benefits and external flexibility revenues;
- a detailed analysis of the interactions between ancillary service provision and REC operation by separately quantifying the main economic and energy-related components affected by demand-side flexibility.

1.3. Limitations of the study

The proposed framework is subject to some limitations that should be considered when interpreting the results.

First, flexibility availability is represented through predefined flexibility levels rather than being derived from an explicit behavioural model of individual users or appliance-level scheduling. Consequently, the analysed flexibility scenarios should be interpreted as sensitivity analyses rather than realistic participation rates. In addition, the proposed framework does not explicitly model user discomfort or behavioural costs associated with load shifting, since flexibility activation is evaluated from the perspective of the REC/BSP rather than individual consumer preferences.

Second, the proposed framework models flexibility exclusively through demand-side load shifting. Other potential flexibility sources, such as battery energy storage systems, electric vehicles, or controllable

renewable generation, are intentionally excluded in order to isolate the contribution of demand flexibility.

Third, the proposed methodology can be readily adapted to different Renewable Energy Community configurations and regulatory frameworks. Nevertheless, the quantitative results, particularly those related to shared energy and the associated economic incentives, depend on the regulatory mechanism adopted. Therefore, the values reported in this work should be interpreted with reference to the current Italian regulatory framework.

Finally, the Balancing Market participation is evaluated assuming exogenous market conditions derived from historical data. The proposed simulator does not model the market clearing process, strategic bidding behaviour, or the interaction among multiple flexibility providers competing for the same ancillary service.

Furthermore, the proposed framework assumes perfect knowledge of renewable generation profiles within the analysed scenarios in order to isolate the impact of demand-side flexibility activation. Therefore, the effect of photovoltaic forecasting errors on the operational performance and the realised balancing revenues is not explicitly quantified. The extension of the framework to stochastic renewable generation scenarios represents a possible direction for future developments.

1.4. Organisation of the paper

The paper is structured as follows. Section 2 outlines the problem framing, while Section 3 presents the corresponding simulation model. Section 4 describes the case study, and Section 5 presents and discusses the results. Finally, Section 7 provides the conclusions.

2. Problem framing

This study proposes a configuration in which an REC participates in the Balancing Market, focusing in particular on trading Replacement Reserve. Unlike conventional generating units, an REC lacks dispatchable generation and has limited upward and downward reserve capacity, restricting its service range primarily to balancing via RR within the same market zone.

In this paper, we therefore assume that flexibility is provided exclusively through load variations carried out by members of the community who are capable of moving fractions of their consumption between two consecutive days: a) by anticipating some shares of the next day's baseline consumption to the present day (*downward* regulation), or b) by delaying some shares of today's baseline consumption to the following day (*upward* regulation). These individual adjustments are aggregated into a single offer submitted by the BSP on behalf of the REC. Meanwhile, the BRP of the community adjusts the baseline power profile of each member for the following day, in order to avoid imbalance penalties.

The REC is assumed to be composed of pure consumers and pure producers, with energy sharing proportional to the degree of contemporaneity of generation and consumption. Storage systems are excluded as flexibility arises solely from demand-side adjustments. All generation assets are assumed to be newly-installed photovoltaic plants below 1 MW, compliant with Legislative Decree 199/21 [30] to qualify for shared energy incentives. Surplus energy physically injected into the public grid is sold to the Gestore dei Servizi Energetici (GSE), according to the Italian regulations in force, receiving the zonal spot energy price ("Ritiro Dedicato", RID). The reward granted to shared energy consists of a fixed component in €/MWh, corresponding to the avoided transmission tariff (TRAS), plus a variable premium tariff (TIP) that depends on the national energy price PUN Index GME (PUN), the plant's size, and the region where the REC is located [10]. The above hypotheses can be summarised as follows:

- The analysis of the EC members' flexibility focuses on programmable loads, basically able to "shift" by some hours a given share of their consumption: forward (if the load is postponed, to presently provide an upward flexibility) or backward (if the load is advanced, to presently provide a downward flexibility). The proposed framework does not explicitly model individual flexible appliances. Instead, flexibility is represented at the aggregate demand level by assuming that a fraction of each member's electrical demand can be temporally shifted, independently of the underlying technologies or end-use devices;
- The BRP can predict the consumption baseline of the community, day by day. After today's provision of flexibility by the BSP, the BRP is also able to correct tomorrow's baseline, in order to avoid penalties corresponding to the interpretation of the second part of the load shift as an imbalance (i.e., the increase/decrease of tomorrow's load, if upward/downward flexibility was provided today);
- The REC is composed of pure producers and pure consumers;
- There are no storage systems in the REC;
- REC's power plants are new and their power is lower than 1 MW;
- Surplus energy is sold to the grid through the RID scheme;
- REC's power plants are only photovoltaic units.

A graphical illustration of the described REC configuration is provided in Fig. 1.

In this framework, flexibility is modelled as a temporal reallocation of electricity consumption between two consecutive days. Specifically, a fraction of the load originally scheduled within a given day may

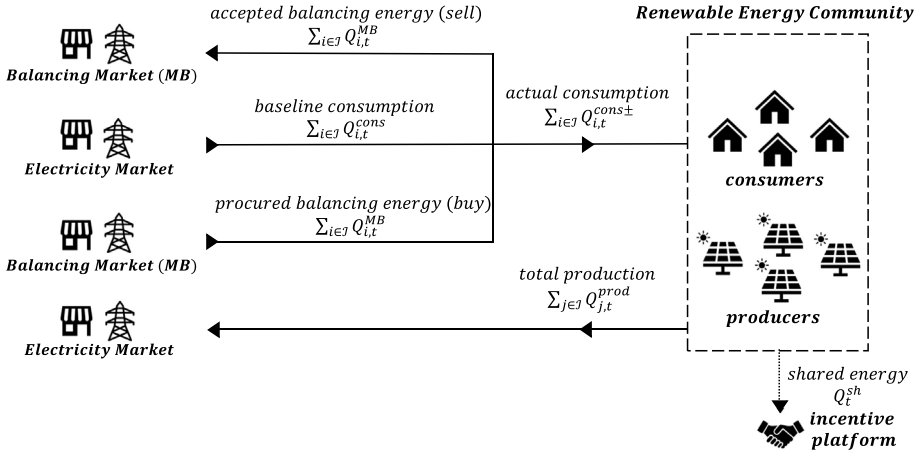


Fig. 1. REC configuration: energy exchanged with the electricity market (EM) and in the balancing market (MB).

be postponed to the following day (upward regulation), or a fraction of the load planned for the following day may be advanced to the current day (downward regulation). This assumption reflects the operational characteristics of programmable residential and commercial loads, whose activation can typically be shifted by several hours but remains constrained within a limited time horizon.

In the proposed formulation, flexibility is therefore exclusively associated with demand-side resources. Non-dispatchable photovoltaic generation is assumed to match the available renewable resource and is not considered a source of flexibility. Consequently, pure producers without electrical demand do not contribute to the flexibility made available to the BSP.

Restricting load shifting to adjacent days is motivated by both regulatory and operational considerations. From a market perspective, balancing actions are activated close to real time, while consumption baselines are defined on a day-ahead basis. Allowing flexibility exchanges only between consecutive days ensures that the Balancing Responsible Party (BRP) has sufficient time to reprogram the community's baseline for the following day after flexibility has been delivered, thereby avoiding imbalance penalties. At the same time, this assumption preserves realism by excluding long-term rescheduling that would be incompatible with household behaviour and current market practices.

The proposed framework is conceived as a simulation tool that reproduces the operational decision-making process of a Balancing Service Provider (BSP) acting on behalf of a Renewable Energy Community. Within this setting, the BSP evaluates, for each market interval, the economic convenience of providing upward or downward regulation in the Balancing Market and determines the corresponding amount of flexible load to be activated, subject to technical constraints of the proposed framework.

Accordingly, flexibility activation is modelled from the perspective of the BSP, which aggregates the individual capabilities of community members and submits a single balancing offer. The allocation of the activated flexibility among individual members is performed according to predefined allocation weights, representing an internal REC management rule. Depending on the adopted organisational framework, the BSP, the REC manager, and the Balance Responsible Party (BRP) may correspond to the same entity or may be operated by different actors. In this framework, the REC does not directly access the MB as an independent market participant; rather, its aggregated flexibility is offered to the MB through the BSP, which acts as the market-facing entity responsible for the balancing offer.

The amount of flexibility available to the BSP is treated as an input parameter of the simulation and represents the share of electrical demand that REC members have made available through predefined technical and contractual agreements. Flexibility levels ranging from 0% to 100% are considered to analyse the impact of different flexibility availability scenarios on REC operation. In particular, the 100% flexibility case represents an extreme-case scenario intended to evaluate the behaviour of the proposed framework under maximum flexibility availability, and should not be interpreted as a realistic participation level or as the result of an explicit behavioural model

3. Simulation model

The proposed model is a static simulator representing the operation of a Renewable Energy Community (REC) participating in the *Mercato di Bilanciamento* (MB). The simulator requires as an input the amounts of load postponement (X) and load anticipation (Y). Given these inputs, along with other technical and economic parameters, it calculates the net energy balance, simulates the REC's market participation, and evaluates the resulting cash flow and other key performance indicators.

Since no empirical data from operational Italian RECs participating in the MB are currently available, the values of X and Y cannot be derived from real-world observations. We therefore propose a set of plausible decision criteria that REC operators could realistically apply in practice under our assumptions to determine these values.

3.1. Model equations and inequalities

The mathematical model governing the simulation is organised as follows: (i) sets and indices, (ii) model parameters, (iii) load shifting dynamics, (iv) feasibility constraints, (v) MB market interactions, and (vi) the economic objective.

3.1.1. Sets and indices

- $\mathcal{T} = \{1, \dots, T\}$: set of all time periods, where T is the total horizon (e.g., number of hourly slots in the year).
- $\mathcal{T}' \subset \mathcal{T}$: first subset of periods where shifting operations are feasible, defined as $\mathcal{T}' := \mathcal{T} \setminus \{1148, (T-48)..T\}$ to exclude initial and final boundary effects.
- $\mathcal{T}'' \subset \mathcal{T}$: second subset of periods where shifting operations are feasible, defined as $\mathcal{T}'' := \mathcal{T} \setminus \{(T-48)..T\}$ to exclude only final boundary effects.
- $\mathcal{EC}$: set of all members (consumers and producers) of the REC.
- $\mathcal{I}$: set of users (consumers) participating in the REC.
- $\mathcal{J}$: set of producers participating in the REC, where $\mathcal{EC} = \mathcal{I} \cup \mathcal{J}$.

3.1.2. Shifting parameters

Load anticipation and postponement are characterised by the following parameters:

- $x_{i,t} \in [0, \eta]$: fraction of the consumption of user i at time t that is shifted forward (postponed).
- $y_{i,t} \in [0, \eta]$: fraction of the consumption of user i at time t that is brought forward (advanced).
- h_t^+: postponement horizon associated with the arrival time t, i.e., the number of periods by which consumption was postponed from its original scheduled time t' to the present time t. Specifically, a quantity arrives at time t if it was originally scheduled at time $t' = t - h_t^+$.
- h_t^-: anticipation horizon associated with the arrival time t, i.e., the number of periods by which consumption is anticipated to the present time t from a future time t''. Specifically, a quantity arrives at time t if it was originally scheduled at time $t'' = t + h_t^-$.
- $\eta \in (0, 1]$: maximum share of baseline consumption of a single time period that can be shifted (either forward or backward), i.e., from 0% to 100%.
- $w_i > 0$: weight associated with user i, representing the relative share of flexibility capacity or contractual priority in the allocation of shifted loads.

X and Y introduced in Section 3 represent the overall amounts of postponed and advanced load, respectively. In the mathematical model, these aggregate quantities are disaggregated into user- and time-specific variables $x_{i,t}$ and $y_{i,t}$, which indicate the share of consumption of user i at period t that is shifted.

The weights w_i act as scaling factors that determine the relative priority of each user in distributing the shifted quantities and are used to compute the allocation quotas. The allocation is performed by the REC manager after the activation of an aggregated flexibility request by the BSP.

The admissible values of the postponement and anticipation horizons associated with each arrival time t are determined based on the position of t within its day. Let each day consist of 24 periods, indexed consecutively starting from period 1. For any $t \in \mathcal{T}$, the corresponding day index is

$$d(t) = \left\lceil \frac{t}{24} \right\rceil,$$

so that $\underline{t}(d)$ and $\bar{t}(d)$, the first and last periods of day $d(t)$, respectively, are given by

$$\underline{t}(d) = 24\,(d(t) - 1) + 1, \qquad \bar{t}(d) = 24\,d(t).$$

Using these definitions, the admissible ranges for the postponement and anticipation horizons are

$$h_t^+ \in \left[t - \left(\left\lfloor \frac{t-1}{24} \right\rfloor \cdot 24\right),\ t - \left(\left\lfloor \frac{t-1}{24} \right\rfloor \cdot 24\right) + 23\right], \quad \forall t \in \mathcal{T}, \tag{1}$$

$$h_t^- \in \left[\left(\left\lceil \frac{t}{24} \right\rceil \cdot 24\right) - t + 1,\ \left(\left\lceil \frac{t}{24} \right\rceil \cdot 24\right) - t + 24\right], \quad \forall t \in \mathcal{T}. \tag{2}$$

The interval for h_t^+ identifies the possible postponement horizons from which consumption originally scheduled earlier within the same day may be shifted to time t. Analogously, the interval for h_t^- identifies the anticipation horizons from which consumption scheduled later within the same day may be advanced to time t. The term $\lfloor (t-1)/24 \rfloor$ identifies the index of the current day when looking backward from t, while $\lceil t/24 \rceil$ identifies the index when looking forward; the resulting bounds enumerate all admissible postponement (h_t^+) and anticipation (h_t^-) distances consistent with the daily structure of the horizon. In other words, at any arrival time t, postponement is possible from any period of the preceding day, and anticipation from any period of the following day.

Numerical examples (hourly data). For example, if $t = 5$ (day 1), the day boundaries are $\underline{t}(1) = 1$ and $\bar{t}(1) = 24$. Therefore:

$$h_5^+ \in [5, 28], \qquad h_5^- \in [20, 43].$$

The corresponding domains of the original scheduling times are:

$$t - h_5^+ \in [\,5 - 28,\ 5 - 5\,] = [-23,\ 0] \text{ (day 0)},$$
$$t + h_5^- \in [\,5 + 20,\ 5 + 43\,] = [25,\ 48] \text{ (day 2)}.$$

However, only the intersections with $\mathcal{T}$ are meaningful, and the sets $\mathcal{T}'$ and $\mathcal{T}''$ are used to ensure that only meaningful periods are retained.

Similarly, when $t = 30$ (day 2), we have $\underline{t}(2) = 25$ and $\bar{t}(2) = 48$. Thus:

$$h_{30}^+ \in [6, 29], \qquad h_{30}^- \in [19, 42].$$

The corresponding domains are:

$$t - h_{30}^+ \in [\,30 - 29,\ 30 - 6\,] = [1,\ 24] \text{ (day 1)},$$
$$t + h_{30}^- \in [\,30 + 19,\ 30 + 42\,] = [49,\ 72] \text{ (day 3)}.$$

Finally, if $t = 48$ (end of day 2), then $\underline{t}(2) = 25$ and $\bar{t}(2) = 48$; we obtain:

$$h_{48}^+ \in [24, 47], \qquad h_{48}^- \in [1, 24].$$

The corresponding domains are:

$$t - h_{48}^+ \in [\,48 - 47,\ 48 - 24\,] = [1,\ 24] \text{ (day 1)},$$
$$t + h_{48}^- \in [\,48 + 1,\ 48 + 24\,] = [49,\ 72] \text{ (day 3)}.$$

3.1.3. Load shifting dynamics

The actual consumption of user i at time t after a shifting corresponds to:

$$Q_{i,t}^{cons\pm} := Q_{i,t}^{cons} \cdot \left(1 - \left((x_{i,t} + y_{i,t})\right)\right) + \left(Q_{i,(t-h_t^+)}^{cons} \cdot x_{i,(t-h_t^+)}\right) + \left(Q_{i,(t+h_t^-)}^{cons} \cdot y_{i,(t+h_t^-)}\right)$$
$$\forall t \in \mathcal{T}' := \mathcal{T} \setminus \{1..48, (T-48)..T\}, \forall i \in \mathcal{I} \tag{3}$$

Here, $Q_{i,t}^{cons}$ is the original baseline consumption, while $Q_{i,t}^{cons\pm}$ represents the adjusted load after considering postponement ($x_{i,t}$) and anticipation ($y_{i,t}$). The variable $x_{i,t}$ denotes the fraction of the load originally scheduled at time t that is postponed to a later time, whereas $y_{i,t}$ denotes the fraction anticipated from t to an earlier time.

Accordingly, the expression $Q_{i,t}^{cons}(1 - (x_{i,t} + y_{i,t}))$ reduces the original baseline load at time t by the portions that are shifted away from t through postponement ($x_{i,t}$) or anticipation ($y_{i,t}$). The term $Q_{i,(t-h_t^+)}^{cons} x_{i,(t-h_t^+)}$ accounts for the load that was originally scheduled at the earlier time $t' = t - h_t^+$ and postponed to t, thus increasing the load at t. Similarly, the term $Q_{i,(t+h_t^-)}^{cons} y_{i,(t+h_t^-)}$ adds the load that was originally scheduled at the later time $t'' = t + h_t^-$ and anticipated to t.

In summary, at each period t, the actual load decreases by the amount postponed from t and by the amount anticipated from t, while it increases by the load arriving from an earlier period via postponement and from a later period via anticipation.

3.1.4. Feasibility constraints

The input parameters $x_{i,t}$ and $y_{i,t}$ must satisfy the following inequalities:

$$x_{i,t} \le b_{i,t} \cdot \eta, \qquad \forall i \in \mathcal{I}, \forall t \in \mathcal{T} \tag{4}$$

$$y_{i,t} \le (1 - b_{i,t}) \cdot \eta, \qquad \forall i \in \mathcal{I}, \forall t \in \mathcal{T} \tag{5}$$

$$\underline{Q}^{cons\pm} \le Q_{i,t}^{cons\pm} \le \overline{Q}^{cons\pm}, \qquad \forall i \in \mathcal{I}, \forall t \in \mathcal{T} \tag{6}$$

$$b_{i,t} \in \{0, 1\}, \qquad \forall i \in \mathcal{I}, \forall t \in \mathcal{T} \tag{7}$$

$$x_{i,t}, y_{i,t} \in [0, \eta], \qquad \forall i \in \mathcal{I}, \forall t \in \mathcal{T} \tag{8}$$

Constraints (4)–(5) guarantee that, in each period, only one shifting direction is possible (either forward or backward) and is limited by the parameter η. Constraint (6) enforces lower and upper bounds on the adjusted load $Q_{i,t}^{cons\pm}$, where $\underline{Q}^{cons\pm}$ and $\overline{Q}^{cons\pm}$ denote its admissible range. The indicator parameter $b_{i,t}$ in (7) encodes the active shifting direction: $b_{i,t} = 1$ corresponds to load postponement ($x_{i,t}$), while $b_{i,t} = 0$ corresponds to load advancement ($y_{i,t}$). Finally, (8) states that $x_{i,t}$ and $y_{i,t}$ are bounded continuous parameters in the interval $[0, \eta]$.

3.1.5. Market interactions

The REC interacts with both internal sharing mechanisms and the external Mercato di Bilanciamento. The amount of energy shared within the REC in each period is defined as:

$$Q_t^{sh} := \min\left(\sum_{i \in \mathcal{I}} Q_{i,t}^{cons\pm},\ \sum_{j \in \mathcal{J}} Q_{j,t}^{prod}\right), \qquad \forall t \in \mathcal{T}' \tag{9}$$

where $Q_{i,t}^{cons\pm}$ is the adjusted consumption of user i after shifting, and $Q_{j,t}^{prod}$ is the production of member j. Eq. (9) ensures that shared energy cannot exceed the smaller of the aggregate demand and supply at time t.

Participation in the MB is modelled as:

$$0 \le Q_{i,t}^{MB+} \le Q_{i,t}^{cons} \cdot x_{i,t}, \qquad \forall i \in \mathcal{I}, \forall t \in \mathcal{T}' \tag{10}$$

$$0 \le Q_{i,t}^{MB-} \le Q_{i,t+h_t^-}^{cons} \cdot y_{i,t+h_t^-}, \qquad \forall i \in \mathcal{I}, \forall t \in \mathcal{T}' \tag{11}$$

where $Q_{i,t}^{MB+}$ denotes the quantity accepted by the MB (sell side), while $Q_{i,t}^{MB-}$ represents the quantity procured from the MB (buy side). Constraints (10)–(11) limit traded quantities to the fractions of baseline consumption that are actually shifted.

At the system level, total MB trades are bounded by:

$$\sum_{i \in \mathcal{I}} Q_{i,t}^{MB+} \le \overline{Q}_t^{MB+}, \qquad \forall t \in \mathcal{T}' \tag{12}$$

$$\sum_{i \in \mathcal{I}} Q_{i,t}^{MB-} \le \overline{Q}_t^{MB-}, \qquad \forall t \in \mathcal{T}' \tag{13}$$

where $\overline{Q}_t^{MB+}$ and $\overline{Q}_t^{MB-}$ are the zonal requests of the balancing market.

3.1.6. Economic function

The economic performance of the REC is measured through the cash flow (CF), which aggregates all monetary inflows and outflows across the time horizon:

$$\begin{aligned}\mathrm{CF} := \sum_{t\in\mathcal{T}'} \Bigg[& \sum_{j\in\mathcal{J}} \pi_t^{ZONE}\, Q_{j,t}^{prod} - \sum_{i\in\mathcal{I}} \pi_t^{PUN}\, Q_{i,t}^{cons} + \pi_t^{sh}\, Q_t^{sh} \\ & + \sum_{i\in\mathcal{I}} \left(\pi_t^{MB+}\, Q_{i,t}^{MB+} - \pi_t^{MB-}\, Q_{i,t}^{MB-} \right) \\ & + \sum_{i\in\mathcal{I}} \pi_t^{PUN} \left(Q_{i,t}^{MB+} - Q_{i,t-h_t^+}^{MB+} \right) \\ & + \sum_{\substack{i\in\mathcal{I} \\ t'\in\mathcal{T}'':\, t'+h_{t'}^-=t}} \pi_t^{PUN}\, Q_{i,t'}^{MB-} \Bigg] \end{aligned} \tag{14}$$

The economic objective in (14) consists of six hourly components, each reflecting a distinct contribution to the community's overall cash flow.

1. The first hourly term represents the revenues obtained by selling the production of community generators at the zonal price; this first term remains constant since no flexibility is activated by the producers;
2. The second term accounts for the expenditure associated with purchasing the baseline consumption profile –that is, the original demand before any flexibility actions—at the PUN price;
3. The third term is the monetary incentive corresponding to energy sharing, valued at π_t^{sh};
4. The fourth term accounts for the benefits obtained by postponing a share of consumption, remunerated at the πt^{MB+} price, as well as the costs incurred when anticipating consumption, purchased at the πt^{MB-} price, as resulting from participating in the MB market;
5. The fifth term describes the correction with respect to the PUN price for the quantities shifted at time t. More precisely, it is positive for the quantities postponed from t, which no longer need to be purchased at PUN price at that instant, and negative for the quantities anticipated at t, which effectively increase the baseline consumption at t;
6. The sixth term represents the cost reduction associated with the energy that has been anticipated from time t to earlier instants t'. These quantities are no longer purchased at the PUN price at time t, as they were already purchased at the πt^{MB-} price at the corresponding anticipation time t'.

The condition

$$t' + h_{t'}^- = t$$

appearing in the last summation selects all previous instants t' for which an anticipation of duration $h_{t'}^-$ shifts a portion of the load, originally scheduled at time t, to the earlier instant t'. Hence, the sum runs over all anticipation events that remove consumption from t, ensuring that the associated PUN savings are correctly attributed to the current time step.

The unitary reward for energy sharing π_t^{sh} is determined as:

$$\pi_t^{sh} := \pi_t^{TIP} + \pi^{TRAS}, \qquad \forall t \in \mathcal{T} \tag{15}$$

where π^{TRAS} is the regulated transmission tariff.

The incentive component π_t^{TIP}, expressed in €/MWh, is calculated according to the current Italian regulation:

$$\pi_t^{TIP} := \min\left(80 + \max\left(0, 180 - \pi_t^{PUN}\right),\ 120\right) + k_z, \qquad \forall t \in \mathcal{T} \tag{16}$$

where k_z is a zonal constant, while π_t^{PUN} is the *Prezzo Unico Nazionale* (PUN, the national spot price).

In order to facilitate readability, Tables 1–3 summarise the notation adopted in the model, reporting respectively the parameters, the hourly energy quantities, and the price components.

Table 1
Model nomenclature: Parameters.

Parameters	Description	Unit
h_t^+	Hours of postponement	hours
h_t^-	Hours of anticipation	hours
η	Maximum flexibility	p.u.
$b_{i,t}$	Binary parameter	{ 0, 1 }
$x_{i,t}$	Provided upward flexibility	p.u.
$y_{i,t}$	Provided downwards flexibility	p.u.
w_i	Weight assigned to user i for flexibility allocation	p.u.

Table 2
Model nomenclature: Hourly energy quantities.

Parameters	Description	Unit
Q_t^{sh}	Energy quantity shared within the REC	MWh
$Q_{i,t}^{MB+}$	Upward flexibility provided to the MB (sell bids) by the single consumer	MWh
$Q_{i,t}^{MB-}$	Downward flexibility provided to the MB (buy bids) by the single consumer	MWh
$Q_{j,t}^{prod}$	Energy quantity supplied by the j-th producer	MWh
$Q_{i,t}^{cons\pm}$	Energy quantity consumed by the i-th consumer including flexibility's operations	MWh
$\underline{Q}^{cons\pm}$	Lower bound of $Q_{i,t}^{cons\pm}$	MWh
$\overline{Q}^{cons\pm}$	Upper bound of $Q_{i,t}^{cons\pm}$	MWh
$\overline{Q}_t^{MB+}$	Upper bound of $Q_{i,t}^{MB+}$	MWh
$\overline{Q}_t^{MB-}$	Upper bound of $Q_{i,t}^{MB-}$	MWh
$Q_{i,t}^{cons}$	Original baseline of the i-th user at time t	MWh

Table 3
Model nomenclature: Prices.

Parameters	Description	Unit
π_t^{ZONE}	Zonal spot price of electricity	€/MWh
π_t^{PUN}	National spot price of electricity (PUN)	€/MWh
π_t^{sh}	Total reward for energy sharing	€/MWh
π_t^{TIP}	Incentive to energy sharing (national part)	€/MWh
k_z	Incentive to energy sharing (zonal part)	€/MWh
π^{TRAS}	Transmission tariff	€/MWh
π_t^{MB+}	Upward balancing price	€/MWh
π_t^{MB-}	Downward balancing price	€/MWh

3.2. Criteria for defining X and Y

As already mentioned, since no empirical data are available for the aggregated postponement (X) and advancement (Y) amounts, these quantities are generated by a rule-based procedure that compares current MB prices with the PUN prices of the entire day following that of flexibility supply (i.e., containing t). Let S denote the number of time periods per day (e.g., $S = 24$ for hourly data). For each $t \in \mathcal{T}'$ we define the set of candidate target periods on the following day as

$$\mathcal{T}_t^{\text{next}} := \{\, t'' \in \mathcal{T} \mid \lfloor (t''-1)/S \rfloor = \lfloor (t-1)/S \rfloor + 1 \,\}. \tag{17}$$

Example (hourly data). If $S = 24$ and $t = 1$, then $\mathcal{T}_t^{\text{next}} = \{25, \ldots, 48\}$; if $t = 2$ then $\mathcal{T}_t^{\text{next}} = \{25, \ldots, 48\}$ as well, and so on—i.e., for every t of day 1 we compare with all 24 candidate hours of the following day. Similarly, if $t = 26$ (day 2), then $\mathcal{T}_{26}^{\text{next}} = \{49, \ldots, 72\}$, i.e., we compare with all 24 candidate hours of the following day (day 3).

Although in the original formulation anticipation was expressed as a shift from t to t', for algorithmic compactness and to unify the treatment of postponement and anticipation, we now simulate all flexibility decisions relative to the current period t. When simulating a flexibility decision at period t, this period serves as the reference point. In the case of postponement, consumption is shifted from t (departure) to t'' (arrival). In the case of anticipation, consumption is shifted from t'' (departure) to t (arrival).

For each origin period t and for each candidate target $t'' \in \mathcal{T}_t^{\text{next}}$, we compute two price differentials:

$$\Delta^{\text{sell}}_{t,t''} := \pi_t^{MB+} - \pi_{t''}^{PUN}, \tag{18}$$

$$\Delta^{\text{buy}}_{t,t''} := \pi_{t''}^{PUN} - \pi_t^{MB-}. \tag{19}$$

Intuitively, $\Delta^{\text{sell}}_{t,t''}$ measures the gain from offering energy in MB at time t (upward regulation) while postponing the corresponding consumption to t'', whereas $\Delta^{\text{buy}}_{t,t''}$ measures the gain from procuring energy in MB at time t (downward regulation) while anticipating the relevant consumption from t''. The flexibility market prices of the current time step t are known, while the knowledge of the PUN prices of the hours t'' of the following day will be discussed in Section 4.1.

Decision rule applied to each $t \in \mathcal{T}'$:

1. Evaluate $\Delta^{\text{sell}}_{t,t''}$ and $\Delta^{\text{buy}}_{t,t''}$ for all $t'' \in \mathcal{T}_t^{\text{next}}$.
2. Find the best candidate.

$$(t^\star, d^\star) \in \arg \max_{t'' \in \mathcal{T}_t^{\text{next}}} \max \left\{ \Delta^{\text{sell}}_{t,t''}, \Delta^{\text{buy}}_{t,t''} \right\},$$

   where $d^\star \in \{\text{sell}, \text{buy}\}$ indicates which of the two deltas attains the maximum.
3. If $\max\{\Delta^{\text{sell}}_{t,t^\star}, \Delta^{\text{buy}}_{t,t^\star}\} \leq 0$ then no shifting is performed for origin t (both options are unprofitable).
4. Otherwise, apply the corresponding shift:
   - if $d^\star$ = sell then postponement is performed, assigning the parameter $x_{i,t}$,
   - if $d^\star$ = buy then an advancement is performed, assigning the parameter $y_{i,t}$.

   The total shifted amount is set to the maximum feasible share up to η, subject to the technical limits and to the availability of the target period (a target period that has already been allocated a shifted consumption cannot accept additional shifted quantities).

Once the aggregate shifting decision is made, the shifted amount must be allocated to individual users in proportion to predefined weights w_i (reflecting, for example, contract rules or flexibility availability) and respecting their residual load margins.

Having defined the decision rules for identifying whether a postponement (X) or advancement (Y) is convenient, we now describe the simulation procedure used to generate the corresponding flexible consumption profiles.

Algorithm 1 provides the main simulation routine. It takes as input the original consumption and production profiles of each user, the market prices ($\pi_t^{MB+}, \pi_t^{MB-}, \pi_t^{PUN}$), the flexibility parameter η, and the set of user weights w_i. For each time period $t \in \mathcal{T}'$, the algorithm evaluates all possible target periods on the following day, computes the price differentials ($\Delta^{sell}, \Delta^{buy}$), and selects the most profitable option. If the best differential is non-positive, no shifting is performed. Otherwise, depending on whether the chosen option corresponds to a "sell" (postponement) (Algorithm 2) or a "buy" (advancement) (Algorithm 3), the algorithms determine the shift horizon, the total feasible shift Q_t^{shift}, and then proceed to allocate this shift among individual users.

The allocation step is non-trivial, since shifted quantities must be distributed proportionally to predefined user quotas while respecting the available consumption margins. To this end, the main algorithm calls the subroutine `allocate_loads`, detailed in Algorithm 4. This function implements an iterative allocation process: at each round, only users with residual availability are considered, their quotas are normalised, and tentative shares are computed. Effective allocations are then obtained by applying the technical limits of each user, and the remaining shift R is updated accordingly. The process continues until either the entire shift is assigned or no further availability remains.

Together, these two algorithms ensure that (i) only profitable shifts are performed, (ii) the global shift never exceeds technical or contractual limits, and (iii) the allocations remain fair and consistent with the user weights w_i.

Algorithm 1: Simulate Flexible Loads $Q_{i,t}^{\text{cons}\pm}$.

Input: $Q_{i,t}^{cons}$, $Q_{i,t}^{prod}$, π_t^{MB+}, π_t^{MB-}, π_t^{PUN}, η, $\mathcal{T}, \mathcal{T}', \mathcal{I}, w_i$
Output: $Q_{i,t}^{cons\pm}$, $x_{i,t}$, $y_{i,t}$

Initialise $Q_{i,t}^{cons\pm} \leftarrow Q_{i,t}^{cons}$, $x_{i,t} \leftarrow 0$, $y_{i,t} \leftarrow 0$
Compute user quotas $q_i = \dfrac{w_i}{\sum_{j \in \mathcal{I}} w_j}, \forall i \in \mathcal{I}$
Sort users by decreasing w_i
foreach $t \in \mathcal{T}'$ **do**
  `// Step 1: Evaluate all possible shifts with the next-day intervals`
  Build candidate set $\mathcal{T}_t^{\text{next}}$ (intervals of the next day).
  foreach $t'' \in \mathcal{T}_t^{next}$ **do**
    Compute $\Delta_{t,t''}^{sell} = \pi_t^{MB+} - \pi_{t''}^{PUN}$
    Compute $\Delta_{t,t''}^{buy} = \pi_{t''}^{PUN} - \pi_t^{MB-}$
  `// Step 2: Select best delta`
  Let $(\Delta^\star, t^\star, d) \leftarrow \max_{t'' \in \mathcal{T}_t^{\text{next}},\, d \in \{\text{sell,buy}\}} \Delta_{t,t''}^{d}$
  if $\Delta^\star \leq 0$ **then**
    `// No shifting; continue to next t`
  `// Step 3: Perform shifting according to best option`
  if $d = sell$ **then**
    $Q_{i,t}^{cons\pm}, x_{i,t}, t^\star \leftarrow$ `postpone_loads`$(Q_{i,t}^{cons}, Q_{i,t}^{cons\pm}, \eta, \overline{Q}_t^{MB+}, t, t^\star, q_i, \mathcal{T}, \mathcal{I})$
  else
    $Q_{i,t}^{cons\pm}, y_{i,t}, t^\star \leftarrow$ `advance_loads`$(Q_{i,t}^{cons}, Q_{i,t}^{cons\pm}, Q_{j,t}^{prod}, \eta, \overline{Q}_t^{MB-}, t, t^\star, q_i, \mathcal{T}, \mathcal{I})$
return $Q_{i,t}^{cons\pm}, x_{i,t}, y_{i,t}$

Algorithm 2: `postpone_loads`$(Q_{i,t}^{cons}, Q_{i,t}^{cons\pm}, \eta, \overline{Q}_t^{MB+}, t, t^\star, q_i, \mathcal{T}, \mathcal{I})$.

Input: $Q_{i,t}^{cons}, Q_{i,t}^{cons\pm}, \eta, \overline{Q}_t^{MB+}, t, t^\star, q_i, \mathcal{T}, \mathcal{I}$
Output: $Q_{i,t}^{cons\pm}$, $x_{i,t}$, $t^\star$

$Q_t^{shift} \leftarrow \min\left(\overline{Q}_t^{MB+}, \left(\eta \cdot \sum_{i \in \mathcal{I}} Q_{i,t}^{cons}\right)\right)$
Initialise $Q_{i,t}^{alloc} \leftarrow 0, \forall i \in \mathcal{I}$
$Q_{i,t}^{alloc} \leftarrow$ `allocate_loads`$(Q_{i,t}^{cons}, q_i, Q_t^{shift}, \mathcal{T}, \mathcal{I})$
foreach $i \in \mathcal{I}$ **do**
  $Q_{i,t}^{cons\pm} \leftarrow Q_{i,t}^{cons\pm} - Q_{i,t}^{alloc}$
  $Q_{i,t^\star}^{cons\pm} \leftarrow Q_{i,t^\star}^{cons\pm} + Q_{i,t}^{alloc}$
  $x_{i,t} \leftarrow \left(\left(Q_{i,t}^{cons} - Q_{i,t}^{cons\pm}\right) / Q_{i,t}^{cons}\right)$
Mark $t^\star$ as used
return $Q_{i,t}^{cons\pm}$, $x_{i,t}$, $t^\star$

3.3. Simulation workflow

The overall simulation process is summarised in the flowchart in Fig. 2. It highlights the main modules of the simulator, from input data and decision rules, to load computation, market interactions, and final outputs.

4. Case study

4.1. Description

To illustrate the features of the proposed methodology, a realistic REC located in the Northern market zone is considered. The REC is located in the Northern Italy market zone, in a coastal area of the Liguria

Algorithm 3: `advance_loads`$(Q^{cons}_{i,t}, Q^{cons\pm}_{i,t}, Q^{prod}_{j,t}, \eta, \overline{Q}^{MB-}_{t}, t, t^\star, q_i, \mathcal{T}, \mathcal{I})$.

Input: $Q^{cons}_{i,t}, Q^{cons\pm}_{i,t}, Q^{prod}_{j,t}, \eta, \overline{Q}^{MB-}_{t}, t, t^\star, q_i, \mathcal{T}, \mathcal{I}$
Output: $Q^{cons\pm}_{i,t}$, $y_{i,t}$, $t^\star$

$Q^{shift}_{t} \leftarrow \min\left(\overline{Q}^{MB-}_{t}, \left(\eta \cdot \sum_{i\in\mathcal{I}} Q^{cons}_{i,t^\star}\right)\right)$
Initialise $Q^{assign}_{i,t} \leftarrow 0, \forall i \in \mathcal{I}$
foreach *user* j *in decreasing order of* w_j **do**
  if $Q^{prod}_{j,t} > 0 \ \wedge \ Q^{cons}_{j,t} > 0$ **then**
    Compute surplus $S_{j,t} \leftarrow \max(Q^{prod}_{j,t} - Q^{cons}_{j,t}, 0)$
    Assigned share $A_{j,t} \leftarrow \min(Q^{shift}_{t}, S_{j,t})$
    Update $Q^{assign}_{j,t} \leftarrow A_{j,t}$
    Update residual shift $Q^{shift}_{t} \leftarrow Q^{shift}_{t} - A_{j,t}$
    if $Q^{shift}_{t} \leq 0$ **then**
      break
if $Q^{shift}_{t} > 0$ **then**
  $Q^{alloc}_{i,t} \leftarrow$ `allocate_loads`$(Q^{cons}_{i,t^\star}, q_i, Q^{shift}_{t}, \mathcal{T}, \mathcal{I})$
foreach $i \in \mathcal{I}$ **do**
  $Q^{cons\pm}_{i,t} \leftarrow Q^{cons\pm}_{i,t} + Q^{alloc}_{i,t} + Q^{assign}_{i,t}$
  $Q^{cons\pm}_{i,t^\star} \leftarrow Q^{cons\pm}_{i,t^\star} - Q^{alloc}_{i,t} - Q^{assign}_{i,t}$
  $y_{i,t^\star} \leftarrow \left(\left(Q^{cons}_{i,t^\star} - Q^{cons\pm}_{i,t^\star}\right)/Q^{cons}_{i,t^\star}\right)$
Mark $t^\star$ as used
return $Q^{cons\pm}_{i,t}$, $y_{i,t}$, $t^\star$

Algorithm 4: `allocate_loads`$(Q^{cons}_{i,t}, q_i, Q^{shift}_{t}, \mathcal{T}, \mathcal{I})$.

Input: $Q^{cons}_{i,t}, q_i, Q^{shift}_{t}, \mathcal{T}, \mathcal{I}$
Output: $Q^{alloc}_{i,t}$ (allocated shift per user)

Initialise $Q^{alloc}_{i,t} \leftarrow 0, \quad A_i \leftarrow Q^{cons}_{i,t}$ (available per user)
Set $R \leftarrow Q^{shift}_{t}$ (remaining shift to allocate)
while $R > 0$ ***and*** $\exists i : A_i > 0$ **do**
  Compute active quotas:

$$q'_i = \begin{cases} q_i & \text{if } A_i > 0 \\ 0 & \text{otherwise} \end{cases}$$

  Normalise quotas: $\hat{q}_i \leftarrow \dfrac{q'_i}{\sum_j q'_j}$
  Tentative allocation: $\Delta_i \leftarrow \hat{q}_i \cdot R$
  Effective allocation: $\delta_i \leftarrow \min(\Delta_i, A_i)$
  Update:

$$Q^{alloc}_{i,t} \leftarrow Q^{alloc}_{i,t} + \delta_i, \quad A_i \leftarrow A_i - \delta_i, \quad R \leftarrow R - \sum_i \delta_i$$

return $Q^{alloc}_{i,t}$

region. It comprises six members: five non-domestic consumers with a total annual consumption of approximately 350 MWh, and one producer operating a single photovoltaic power plant. The simplicity of the adopted generation scenario does not limit the significance of this case study, as the focus of this work is on the flexibility contribution provided by the passive users of the community.

Two versions of the case study are implemented for each scenario, differing in PV capacity: 15 kWp and 150 kWp, respectively. The latter represents an intentionally oversized installation, introduced to better assess the potential impact of local generation on flexibility provision.

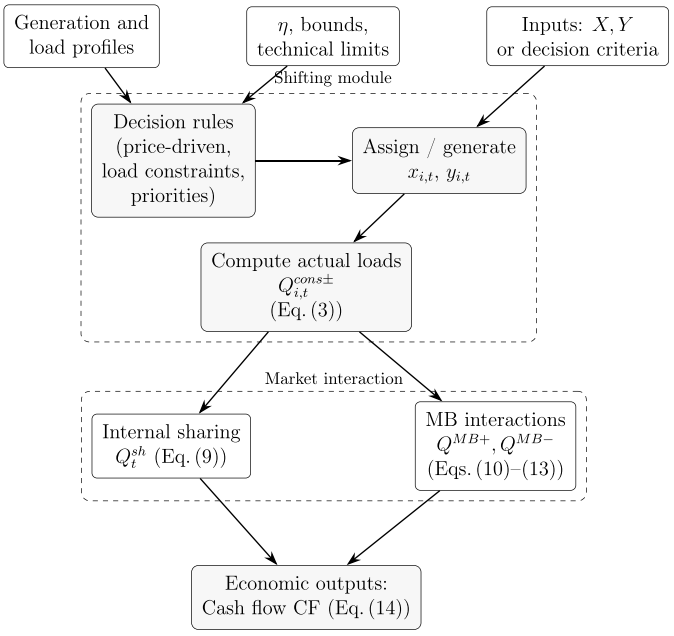


Fig. 2. Flowchart of the simulator: inputs, shifting decisions, load computation, market interactions and outputs.

The proposed methodology is implemented by the BSP through a rolling-horizon simulation. Decisions are updated on an hourly basis: at each hour t, the simulation is performed considering an operational time window extending from the current hour up to the midnight of the following day. Decisions referring to past hours simulations are considered irrevocable, while only future actions can be adjusted.

Within this framework, at each decision step the BSP can either postpone the load of the current hour to any of the 24 hours of the following day (if capacity is still available), or anticipate the load of a future hour from the following day to the current hour, also respecting availability constraints.

As a consequence, the effective simulation horizon is not fixed at $t + 24$, but evolves dynamically throughout the day, refreshing the time window every hour and resetting it at the beginning of each new day. This approach allows capturing the intra-day variability of prices and PV generation while maintaining a realistic decision-making process.

Regarding price information, two operational scenarios are considered:

- **Scenario 1:** The BSP has perfect foresight of all relevant prices (π^{zone}_t, π^{PUN}_t, $\pi^{MB\pm}_t$) for the current day and for the following 24 hours.
- **Scenario 2:** A more realistic assumption is adopted. If the algorithm is executed before 1:00 PM, the BSP only knows the prices (π^{zone}_t, π^{PUN}_t, $\pi^{MB\pm}_t$) of the remaining hours of the current day. Therefore, during the first 13 hours, decisions are made assuming that tomorrow's prices are equal to today's.

It is worth noting that Scenario 2 provides a more realistic setting than Scenario 1, where decisions benefit from perfect foresight, resulting in more targeted and cost-effective load shifting.

The analysis includes four cases combining scenarios and production sizes:

- Case 1.A: Scenario 1 with a 15-kWp PV power plant
- Case 1.B: Scenario 1 with a 150-kWp PV power plant
- Case 2.A: Scenario 2 with a 15-kWp PV power plant
- Case 2.B: Scenario 2 with a 150-kWp PV power plant

Table 4
Aggregated consumption indicators of REC members. Values are anonymised and rounded for readability. Hourly energy consumption and peaks are reported in kWh, while annual energy consumption is in MWh.

Member	Avg. Hourly Consumption [kWh]	Peak Hourly Consumption [kWh]	Annual Consumption [MWh]
Member1	29.9	110.5	262.3
Member2	0.3	3.5	3.0
Member3	2.9	25.1	25.5
Member4	5.5	36.5	47.8
Member5	0.9	7.5	7.9
Member6	0.0	0.0	0.0
Total	39.5	118.2	346.4

Table 5
Relative weights of REC members for flexibility allocation. Member6 is a producer and does not provide load flexibility.

Member	Member1	Member2	Member3	Member4	Member5	Member6
Type	Consumer	Consumer	Consumer	Consumer	Consumer	Producer
Weight	0.5	1	0.6	0.5	0.8	0

4.2. Input data

As discussed in Section 1.2.2, the distinction between FCR, aFRR, and RR is crucial, as the available dataset of prices and volumes clearly differentiates "secondary reserve" (corresponding to aFRR) from other ancillary services. In this analysis, we focus exclusively on reserve services with continuous remuneration mechanisms, explicitly excluding those compensated through one-off payments or availability fees only. These data are available from GSE and consist of actual awarded prices and hourly quantities exchanged in the MB. The simulation refers to the year 2023. The dataset provides the following information:

- $Q_{i,t}^{MB+}$: energy quantity exchanged for upward regulation [MWh]
- $Q_{i,t}^{MB-}$: energy quantity exchanged for downward regulation [MWh]
- π_t^{MB+}: price of $Q_{i,t}^{MB+}$ [€/MWh]
- π_t^{MB-}: price of $Q_{i,t}^{MB-}$ [€/MWh]
- π_{PUN}: national spot price of purchased electricity [€/MWh]
- π_{ZONE}: zonal spot price of sold electricity [€/MWh]

In addition, for each user and producer, hourly consumption and production data are provided. The data refer to a primary station supplying six members that could potentially form a REC. Consumption data have an hourly granularity over the year 2023 and include different types of users, such as a museum, a residential apartment, a public building, a primary school, and a senior citizens' center.

Some detailed consumption data are not reported here to preserve the privacy of individual members (users and producers). Instead, Table 4 provides aggregated and anonymised indicators for each community member.

Inside the REC, Member1 exhibits the highest average hourly energy consumption, followed by Member4 and Member3. In terms of annual electricity consumption, Member1 is also the largest consumer, followed by Member4, Member3, Member5, and Member2. Member6 acts exclusively as a producer and does not consume electricity.

Table 4 summarizes the aggregated energy consumption indicators of the REC members. The community is characterised by a highly heterogeneous demand profile, with Member1 accounting for the largest share of electricity consumption, followed by Member4 and Member3, while the remaining members exhibit comparatively low or negligible demand.

Table 5 reports the relative weights assigned to each community member for flexibility allocation.

The PV production is simulated based on a future photovoltaic installation, currently under construction. Hourly generation profiles are obtained using location-specific solar irradiance data consistent with the geographic coordinates of the study area, ensuring realistic and reproducible operating conditions. Generation is computed using typical performance assumptions for crystalline silicon PV systems operating in Northern Italy, including standard losses and conversion efficiencies. The installed capacities of 15 kWp and 150 kWp are assumed to scale linearly, while all other technical characteristics remain unchanged. The PV production data were generated by a third-party entity and provided to the REC; therefore, the raw datasets are not publicly available.

The dataset captures seasonal variations in energy use across different consumer categories within the REC. The largest consumer (Member1) and one medium-sized consumer (Member5) exhibit higher electricity demand during the summer months, while another medium-sized consumer (Member4) shows a clear reduction in consumption over the same period. The remaining members (Member2 and Member3) display increasing demand starting from the summer months onward.

During peak periods, i.e., the hours of the year with the highest energy consumption, Member1 reaches an hourly energy demand of approximately 111 kWh, followed by Member4 at around 36 kWh and Member3 at about 25 kWh. Lower peak demands are observed for Member5 (up to 7.5 kWh/h) and Member2 (up to 3.5 kWh/h).

All buildings are located within a radius of approximately 1 km, which makes them well suited for local energy exchange within a community-based framework.

In terms of daily consumption profiles, Member1 presents relatively stable daytime consumption with minor intra-day fluctuations. Member4 shows a distinct morning peak around 10:00 a.m., while Member2 displays a less predictable pattern, typically characterised by morning and evening peaks. Member3 and Member5 exhibit variable daily peaks occurring either in the morning or in the evening, depending on usage patterns.

4.3. Flexible load simulation

The REC can receive revenue by engaging in baseline modifications—either increasing or decreasing its consumption—to support grid balancing. The TSO selects the offers based on an economic criterion, within a market governed by a pay-as-bid mechanism.

In the case of an upward bid, the consumer contributes to upward flexibility by postponing a portion of scheduled energy consumption to the following day, thereby reducing consumption in the current time slot. Conversely, with a downward bid, the consumer provides downward flexibility by advancing to the current day a portion of energy consumption originally planned for the next day, thus increasing consumption in the present day.

As introduced in Section 3, the extent of load shifting is controlled by the flexibility parameter $\eta \in (0, 1]$, which represents the maximum share of baseline consumption that can be reallocated in time. While values of η close to one correspond to a fully flexible demand and are clearly not attainable in practice, the full range of η is explored to assess, from a theoretical standpoint, the economic attractiveness of flexibility provision and to identify the sensitivity of the results to increasing levels of demand-side responsiveness.

4.4. Model implementation

The model has been implemented in a Python environment. All simulations were executed on an AMD Ryzen 9 6900HX processor clocked at 3.30 GHz with 16 GB of RAM.

5. Results

5.1. Profit analysis

The first output considered in the first scenario is the net economic benefit, as shown in Fig. 3 for Case 1.A and in Fig. 4 for Case 1.B. On the

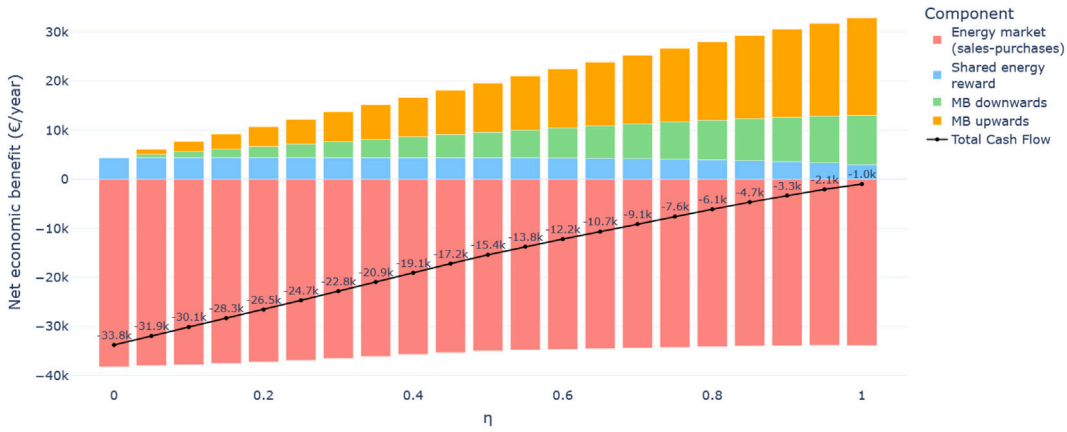


Fig. 3. Profit measurement in case 1.A.

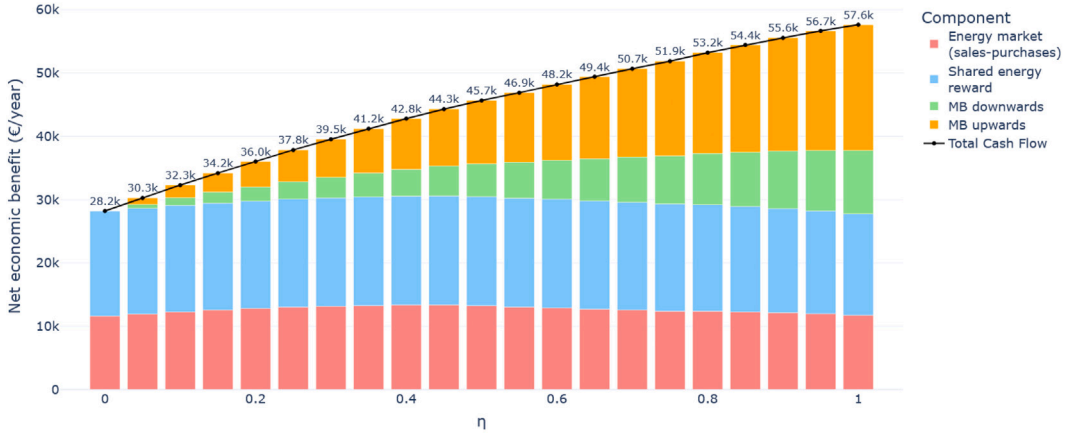


Fig. 4. Profit measurement in case 1.B.

x-axis, the flexibility percentage is reported, ranging from 0 to 100% in steps of 5%, while the y-axis represents the annual net economic benefit, expressed in thousands of euros (k€).

In Fig. 3, we observe that the yearly net balance of revenues and purchases on the energy market, in pink, is always negative; this strongly affects the sign of the total cash flow, despite the presence of shared energy incentives (blue bar), revenues from downward market balancing, MB downwards (green bar), and avoided costs from upward market balancing, MB upwards (orange bar). This is because the amount of energy sold is significantly lower than the amount purchased. In the reference case ($\eta = 0$), without any flexibility, the total yearly net cost on the energy market amounts to €38,212; then, the total cash flow benefits from the shared energy reward (€4435, approximately 12%). With the maximum flexibility value, the net balance of revenues and purchases on the energy market is still €33,884, but the total cash flow significantly increases to –972€. This is achieved thanks to the shared energy reward (€2994, around 10%) and, more substantially, to €29,918 gained in the balancing market (€19,883 and €10,035 from upward and downward flexibility, respectively). Overall, comparing the reference case to the one with maximum flexibility, the total expenditure is reduced by up to 97%.

In contrast, Fig. 4 shows that—since the production is much higher than in the previous case—the energy community achieves a net positive cash flow even in the reference case, with net earnings of €11,630 on the energy market; these are further increased by shared energy incentives of about €16,560. With the maximum level of flexibility, the total yearly cash flow rises to €57,640. This is the result of a net cash flow of €11,757 on the energy market (about 20%), of €15,965 obtained from shared energy incentives (around 30%), and of €29,918 coming from the MB (around 50%), the same as before. Therefore, compared to the reference case without flexibility, net earnings are more than doubled.

Since Cases 2.A and 2.B yield similar results to those already described, only a summary table, which can be consulted in Table 6.

From the values presented in Table 6, we observe that there are only minor differences between Scenario 1 and Scenario 2. Scenario 1 appears slightly more advantageous—as expected—since all prices are known in advance, allowing for more informed load shifting decisions.

For Case A, the contribution of market balancing mechanisms (MB downward revenues and MB upward avoided costs) to the net economic benefit progressively increases with flexibility, becoming comparable to or larger than the shared energy reward at relatively low flexibility values (around $\eta = 15\%$). In contrast, in Case B the shared energy reward remains the dominant contributor to the net economic benefit over a wider range of flexibility levels (up to approximately $\eta = 60\%$). In all cases, the shared energy reward provides a relatively stable contribution across the entire flexibility range, while the impact of MB grows with increasing η.

5.2. Shared energy

The second output considered in the first scenario is the shared energy percentage, as shown in Fig. 5 for Case 1.A and in Fig. 6 for Case 1.B. The shared energy percentage is defined as the ratio between the total shared energy and the total renewable energy production over the

Table 6
Total cash flow and its components for each case at different flexibility levels.

Scenario	Case	η %	Total Cash Flow [e]	Energy market [e]	Shared energy reward [e]	MB upwards [e]	MB downwards [e]
1	1.A	0	–33,777	–38,212	4435	0	0
1	1.A	100	–972	–33,884	2994	19,883	10,035
1	1.B	0	28,190	11,630	16,560	0	0
1	1.B	100	57,640	11,757	15,965	19,883	10,035
2	2.A	0	–33,777	–38,212	4435	0	0
2	2.A	100	–1156	–34,390	3056	9934	19,845
2	2.B	0	28,190	11,630	16,560	0	0
2	2.B	100	57,493	11,590	16,124	9934	19,845

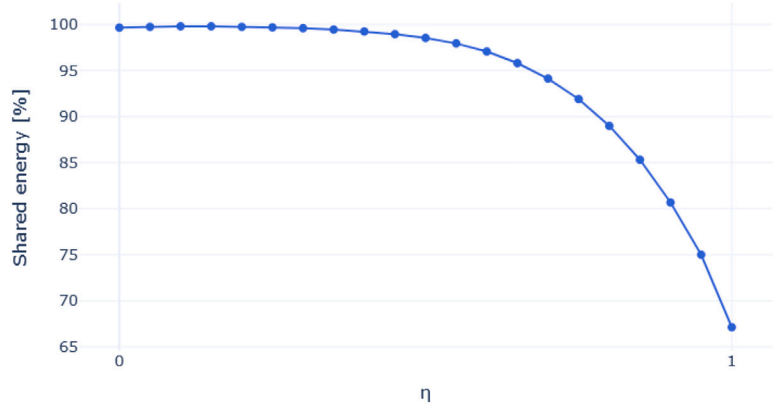


Fig. 5. Shared energy in Case 1.A, as a percentage of REC's generation.

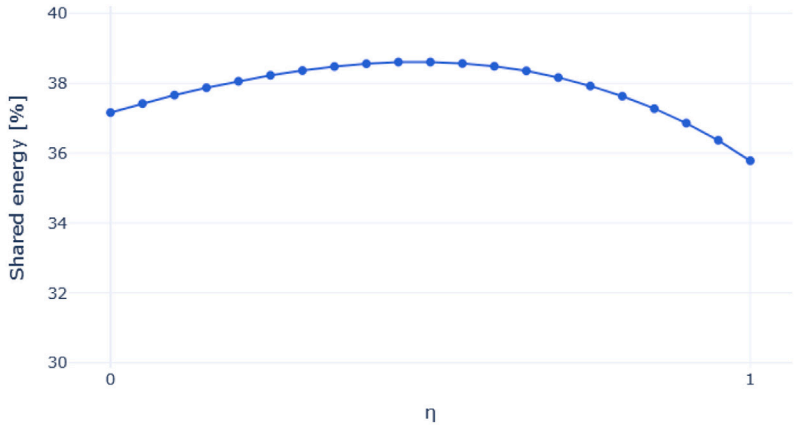


Fig. 6. Shared energy in Case 1.B, as a percentage of REC's generation.

considered time horizon:

$$\text{SharedEnergy}\% = \frac{\sum_{t\in\mathcal{T}} Q_t^{\text{sh}}}{\sum_{t\in\mathcal{T}} \sum_{j\in\mathcal{J}} Q_{j,t}^{\text{prod}}} \times 100$$

On the x-axis, the flexibility percentage is reported, ranging from 0 to 100% with a step of 5%, while the y-axis represents the percentage of shared energy. In Fig. 5, a decreasing trend can be observed. In the reference case, the shared energy is approximately 99.6%, while for the maximum flexibility value it decreases to around 67.1%. The highest value, about 99.8%, is reached at a flexibility level of 10%, while energy sharing remains practically constant throughout the left part of the figure.

This behaviour can be explained by the fact that in Case 1.A the renewable electricity generation is generally insufficient to fully satisfy the community demand. As a consequence, at least for low levels of flexibility, the hourly load remains higher than generation, and the shared energy corresponds to production and is practically independent of the amount of demand response. Only in the few hours with a generation surplus, can a small amount of flexibility increase the load and thereby energy sharing, which justifies the slight peak at $\eta = 10\%$. Nevertheless, for high flexibility levels, the resulting load reallocation becomes detrimental, leading to a decrease in shared energy. Fig. 6 shows a similar trend, but at lower shared energy levels. In this case, the reference shared energy percentage is around 37.2%, decreasing to approximately 35.8% at the maximum flexibility level. The maximum value, around 38.6%, is achieved at a flexibility level of about 50%. This lower level of energy sharing reflects the fact that, in Case 1.B, renewable production is significantly higher than community demand, and therefore only a

Table 7
Shared energy, in percent of generation, for each case at different flexibility levels.

Scenario	Case	η %	Shared Energy %
1	1.A	0	99.6
1	**1.A**	**10**	**99.8 (max)**
1	1.A	100	67.1
1	1.B	0	37.2
1	**1.B**	**50**	**38.6 (max)**
1	1.B	100	35.8
2	2.A	0	99.6
2	**2.A**	**10**	**99.7 (max)**
2	2.A	100	68.5
2	2.B	0	37.1
2	**2.B**	**50**	**38.7 (max)**
2	2.B	100	36.1

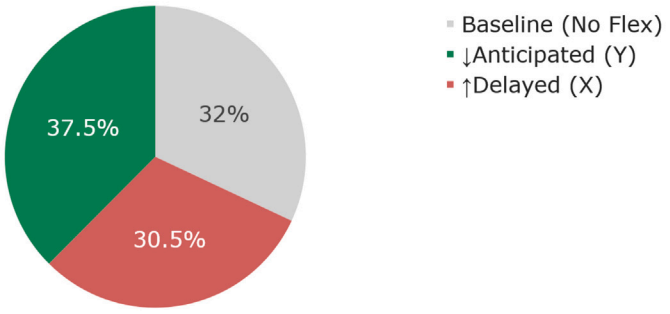


Fig. 7. Shift distribution in case 1.A, for $\eta = 100\%$.

fraction of the generated electricity can be shared, even in the presence of flexibility.

Since Cases 2.A and 2.B produce similar results to those already described, we provide only a summary table, which can be consulted in Table 7.

Overall, in all cases the shared energy percentage exhibits a non-monotonic dependence on the flexibility parameter η, with a maximum attained at an intermediate flexibility level (10% in Cases A and 50% in Cases B). Beyond this point, the shared energy percentage decreases as η approaches its maximum value, with variations ranging from approximately 2.5% in Cases B to more than 32% in Cases A.

5.3. Shift distribution

The third output considered in our simulations is the shift distribution per user, as shown in Fig. 7 for Case 1.A. Specifically, for each user, the distribution of occurrences (hours of the year) when energy has been delayed, anticipated, or left unchanged is reported. Since these percentages are identical across all users, only the values for Member1 are presented. In Table 8 we can also observe that there is no difference between Case 1.A and Case 1.B, since they differ only in generation assets, which are not involved in flexibility. Similarly, in Scenario 2, the results of Case 2.A and Case 2.B are identical; furthermore, they differ only marginally from Scenario 1, due to the low impact of differences in the forecasting accuracy of the energy prices for the following day.

These results provide insights relevant to the conditions under which it was more convenient for the REC to trade on the MB, specifically whether it was more advantageous to submit a sell bid, a buy bid, or if it was not profitable at all, in which case the REC took no action. It is observed that, overall, the decisions remain relatively balanced; however, there seems to be a tendency to favour load anticipation. Although

Table 8
Hours of participation in the balancing market, for $\eta = 100\%$.

Scenario	Case	Delayed %	Anticipated %	Neutral %
1	1.A	30.5	37.5	32
1	1.B	30.5	37.5	32
2	2.A	30.6	37.6	31.8
2	2.B	30.6	37.6	31.8

MB demand generally shows a good balance between sell and buy offers throughout the year, anticipating the load seems to be the more advantageous strategy.

5.4. Flexibility amount

The fourth output considered in the first scenario is the total energy quantity shifted by the community in correspondence with the maximum value of flexibility ($\eta = 100\%$). For Case 1.A, the results are shown in Fig. 8 for sell bids (upward regulation) and in Fig. 9 for buy bids (downward regulation). On the x-axis, the days of the analysed year are reported, while the y-axis represents the daily energy traded for balancing purposes, in kWh. It is worth noting that, in this case, there are some notable differences across cases and scenarios; however, the main findings can still be described based on the figures presented.

It can be observed that, since the shifted load depends on each user's baseline consumption, the yearly trend reflects the consumption characteristics of the users. For instance Member1, who has a higher energy usage in summer, experienced a significant shift in load during that season.

At the maximum flexibility level ($\eta = 100\%$), the total energy shifted by the community over the entire year amounts to approximately 104 MWh for upward regulation and 117 MWh for downward regulation, reflecting the aggregated contribution of all community members.

6. Discussion

The results provide several insights into the interaction between energy sharing within a REC and participation in the MB. The first relevant finding is that the two mechanisms do not necessarily represent competing sources of economic benefit. As flexibility increases, the contribution of MB participation becomes progressively more relevant, whereas the shared energy reward remains relatively stable over a wide range of flexibility levels. Consequently, the reduction in shared energy observed at high flexibility levels can be compensated by the additional revenues obtained from the MB. This indicates that the economic value of flexibility should be assessed by considering both mechanisms jointly rather than evaluating energy sharing and balancing-market participation independently.

The effect of flexibility on energy sharing depends strongly on the balance between renewable generation and community demand. In Case A, where renewable generation is generally insufficient to satisfy community demand, the shared energy percentage remains close to the generation level at low and moderate flexibility levels. Only when flexibility becomes sufficiently high does load reallocation significantly modify the temporal matching between generation and demand, resulting in a marked reduction in shared energy. Conversely, in Case B, where renewable generation substantially exceeds community demand, the amount of energy that can be shared is primarily limited by consumption. As a result, changes in flexibility have a much smaller effect on the shared energy percentage. This suggests that the generation-to-demand ratio of the REC is a key factor in determining the extent to which flexibility provision affects energy sharing.

An additional relevant finding is the non-monotonic relationship between flexibility and shared energy. In both cases, the highest shared energy level is achieved at an intermediate value of η rather than in the absence of flexibility. This behaviour indicates that load

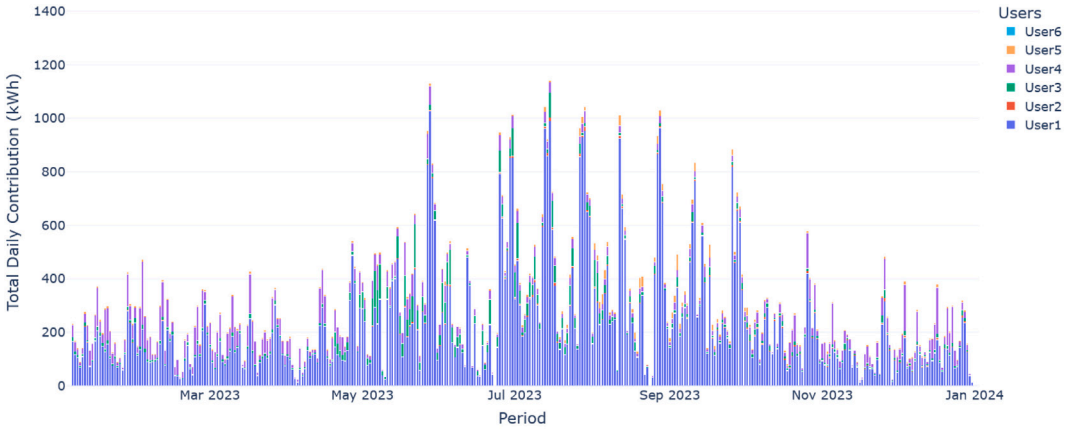


Fig. 8. Flexibility amount in MB market (sell Bid) in case 1.A, for $\eta = 100\%$.

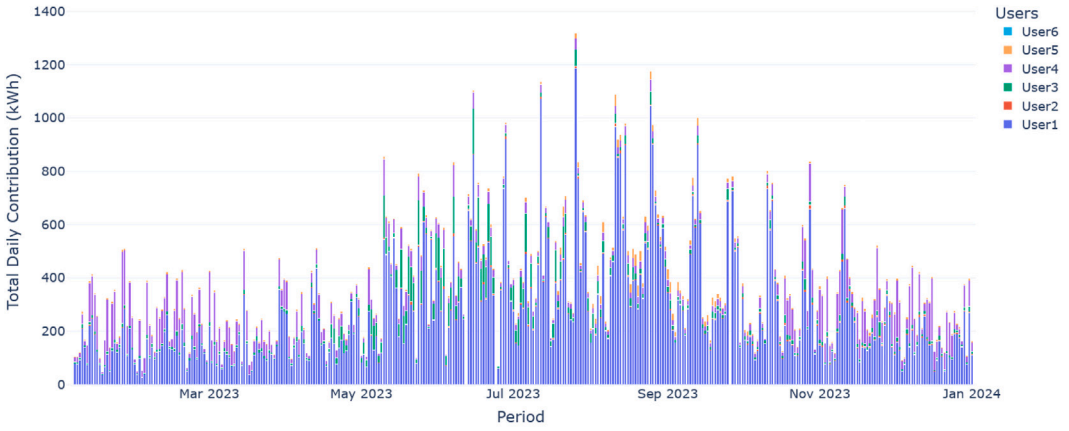


Fig. 9. Flexibility amount in MB market (buy Bid) in case 1.A, for $\eta = 100\%$.

shifting can either improve or deteriorate the temporal alignment between renewable generation and consumption, depending on when the flexibility is activated. At moderate flexibility levels, some load shifts can therefore increase the amount of renewable generation consumed within the community. When flexibility becomes excessive, however, additional load reallocation can move consumption away from periods of renewable generation, reducing shared energy. Flexibility should therefore not be regarded as inherently detrimental to energy sharing; its effect depends on how the available flexibility interacts with the generation profile and the community demand.

The economic results also indicate that the contribution of MB participation becomes relevant already at relatively low flexibility levels, while the total economic performance continues to improve as flexibility increases. This suggests that the economic value of flexibility does not arise only when a large fraction of the demand is made available to the MB.

The amount of shifted energy further illustrates the potential relevance of aggregated REC flexibility. At the maximum flexibility level, the analysed community provides approximately 104 MWh of upward regulation and 117 MWh of downward regulation over the year. Although these quantities are limited when compared with the scale of the overall electricity system, they are relevant at the scale of the analysed community. More importantly, aggregation across multiple users and potentially across multiple communities could substantially increase the amount of flexibility available to the balancing market. This is particularly relevant for REC participation, as an individual community may not always be able to provide sufficient flexibility to meet the technical requirements for market participation.

The comparison between the two price-information scenarios provides an additional result. Only minor differences are observed between the scenario based on perfect knowledge of the following-day prices and the scenario in which this information is approximated using the available data. Under the historical market conditions considered in this study and given the adopted decision-making procedure, perfect knowledge of the following-day PUN therefore provides only a limited additional benefit. This result is relevant because it suggests that the effectiveness of the proposed flexibility strategy may not require highly accurate forecasts of all market variables. Nevertheless, this conclusion should be interpreted within the deterministic framework adopted in this study and should not be extended to balancing-market price uncertainty in general.

The findings are consistent with the broader literature showing that both energy sharing and flexibility provision can generate economic value for distributed energy resources. However, they also highlight the importance of considering their interaction. Studies on REC energy sharing generally focus on increasing self-consumption and shared energy, whereas studies on flexibility markets primarily investigate the provision of ancillary services by aggregated distributed resources. The present results show that demand-side flexibility can modify the amount of energy shared within the REC, but that this effect does not necessarily offset the additional value generated through MB participation. An integrated assessment is therefore important when evaluating the overall economic value of flexibility in a REC.

Table 9
Summary of variation in total cash flow and shared energy, as a percentage of generation, at selected flexibility levels η (Case Study 1.A).

Total Cash Flow	η %	% Shared Energy
–33,777	0.0	99.6
–28,333	15.0	**99.8**
–24,695	25.0	99.6
–20,931	35.0	99.4
–15,405	50.0	98.5
–7624	75.0	91.9
–972	100.0	67.1

From a practical perspective, the proposed simulation framework can support the assessment of different flexibility-availability scenarios before implementing a flexibility programme within a REC. In particular, it enables the economic value of flexibility to be evaluated jointly with its effect on shared energy. This can support REC managers in identifying an appropriate level of flexibility to be made available by members, taking into account both the expected economic benefits and the characteristics of the community's renewable generation and demand profiles.

Overall, the results indicate that participation in the MB can complement rather than undermine the economic value of energy sharing within a REC. The possibility of obtaining additional flexibility revenues while retaining a substantial share of the benefits associated with energy sharing suggests that the joint participation in energy-sharing and balancing mechanisms can represent a promising opportunity to increase the economic value of REC participation in emerging flexibility markets.

7. Conclusions

The obtained results show that the participation of RECs in flexibility markets is profitable, with higher economic returns observed as flexibility increases.

Since the contribution of the shared energy reward is not negligible in the economics of a REC, one of the main targets of this work was understanding whether such contribution was possibly jeopardised by the participation in flexibility markets. We observed that, even if energy sharing varies by some percentage with the amount of provided flexibility, such variations have only a minor impact on the total economics of the community. This is a positive and not an obvious result, as it could have been possible for the shared energy share to vary more significantly—or even decrease substantially—leading to a much lower economic benefit.

Indeed, energy sharing reductions are largely compensated by the earnings from participation in the MB. This can be explained by the fact that, as flexibility increases, its primary impact is on balancing market earnings rather than on shared energy levels; furthermore, even moderate flexibility levels already yield economic benefits that are competitive with the remuneration from shared energy.

Interestingly, for certain values of the flexibility rate, an increase in shared energy is also observed. Therefore, the two mechanisms do not always conflict; in some cases, increasing flexibility can also enhance the remuneration from shared energy, resulting in a double economic benefit.

These insights are further clarified in the summary provided in Table 9, which refers to Case Study 1.A. The results show promising economic returns even with a small percentage of flexibility, suggesting that the required effort or discomfort requested of users is profitable and well rewarded. While the highest benefits are achieved at maximum flexibility levels, these may be less feasible depending on the types of users involved in the REC.

It can also be noticed that these findings are coherent with the fact that the community shifts at most 1.2 MWh per day, which is negligible in absolute terms; however, this is due to the small size of the community. Nevertheless, considering the case with $\eta = 100\%$, where it is optimal to shift approximately two-thirds of the load, as shown in Fig. 7, the result can be considered meaningful. When this fraction is projected onto a national-scale horizon, it clearly highlights the potential of flexibility.

It is worth noting that this study considers a single REC, which could potentially aggregate with other RECs to reach at least the 1 MW threshold required for participation in this type of market.

This work represents a first step in exploring the potential of emerging flexibility markets and the new role that RECs may play in the national energy landscape.

Further steps will include flexibility from generation sources and Battery Energy Storage Systems (BESS). Optimisation techniques will be applied to identify the most profitable strategies for REC participation in the MB.

Finally, for real-world implementation, baseline consumption forecasting, along with evolving assumptions about BRP interactions, and the structure of penalties and rewards will be incorporated.

CRediT authorship contribution statement

Dimitri Thomopulos: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ester Adduci:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Luisa Livia Pia Lambertino:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Data curation, Conceptualization. **Luca Del Greco:** Writing – review & editing, Writing – original draft, Methodology. **Marco Raugi:** Supervision, Project administration, Methodology, Conceptualization. **Davide Poli:** Writing – review & editing, Writing – original draft, Validation, Supervision, Project administration, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

References

[1] European Union. Regulation (EU) 2021/1119 of the European parliament and of the council of 30 June 2021 establishing the framework for achieving climate neutrality and amending regulations (EC) no 401/2009 and (EU) 2018/1999 (European climate law). Official Journal of the European Union, L 243, 9.7.2021, pp. 2021;1–17. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32021R1119 [Accessed: 6 August 2025].
[2] European Parliament. Fit for 55 package; 2024 https://www.europarl.europa.eu/RegData/etudes/BRIE/2022/733513/EPRS_BRI(2022)733513_EN.pdf [Accessed: 21 January 2026].
[3] Commission E. Clean energy for all Europeans package; 2019 https://energy.ec.europa.eu/topics/energy-strategy/clean-energy-all-europeans-package_en [Accessed: 20 February 2024].
[4] Italian Regulatory Authority for Energy Networks and Environment (ARERA). Integrated text of the electricity market rules (TIDE); 2024 https://www.arera.it/fileadmin/allegati/docs/24/539-24rt.pdf [Accessed: 25 June 2025].
[5] Rubino A, Facchini A, Fell M, Giulietti M, Morone A. New energy downstream. Emerging business models and innovative best practices: an economic, institutional, and behavioral focus. Energy Sources, Part B: Economics, Planning, and Policy 2021;16:971–5. https://doi.org/10.1080/15567249.2021.2018647
[6] Casolino G, Del Greco L, Losi A. Flexibility procurement for local ancillary services and provider aggregation perimeter in radial distribution systems. IEEE Access 2024;12:157383–96. https://doi.org/10.1109/ACCESS.2024.3484675
[7] Chen F, Huang C, Wang L, Zhu C, Wang C, Xie N. Flexibility evaluation of distribution network with high penetration of variable generations. In: 2017 IEEE conference on energy internet and energy system integration (EI2); 2017. p. 1–6. https://doi.org/10.1109/EI2.2017.8245479
[8] Khederzadeh M. Defining and realizing flexibility in distribution grid. In: CIRED workshop 2016; 2016. p. 1–4. https://doi.org/10.1049/cp.2016.0767
[9] European Union. Directive (EU) 2019/944 of the European parliament and of the council of 5 June 2019 on common rules for the internal market for electricity and amending directive 2012/27/EU (recast). Official Journal of the European Union,

L 158, 14.6.2019, pp. 2019;125–99. https://eur-lex.europa.eu/legal-content/EN/TXT/?uri=CELEX:32019L0944 [Accessed: 11 June 2025].
[10] Gestore dei Servizi Energetici (GSE). Operational rules for access to the service for diffuse self-consumption and the PNRR contribution; 2024, Approved by the Italian Ministry of Environment and Energy Security, https://www.gse.it/documenti_site/Documenti%20GSE/Servizi%20per%20te/AUTOCONSUMO/Gruppi%20di%20autoconsumatori%20e%20comunita%20di%20energia%20rinnovabile/Regole%20e%20procedure/ALLEGATO%201%20Regole%20operative%20CACER.pdf.
[11] Mariuzzo I, Fioriti D, Guerrazzi E, Thomopulos D, Raugi M. Multi-objective planning method for renewable energy communities with economic, environmental and social goals. Int J Electr Power Energy Syst 2023;153:109331. https://doi.org/10.1016/j.ijepes.2023.109331. https://www.sciencedirect.com/science/article/pii/S0142061523003885.
[12] Trevisan R, Ruggeri S, Ghiani E, Pilo F. Fostering end users' flexibility in renewable energy communities. In: 2023 58th international universities power engineering conference (UPEC); 2023. p. 1–6. https://doi.org/10.1109/UPEC57427.2023.10294918
[13] Mota B, Faria P, Vale Z. Energy cost optimization through load shifting in a photovoltaic energy-sharing household community. Renew Energy 2024;221:119812. https://doi.org/10.1016/j.renene.2023.119812. https://www.sciencedirect.com/science/article/pii/S0960148123017275.
[14] Rollo A, Serafini P, Aleotti F, Cilio D, Morandini E, Moneta D, Rossi M, Zulianello M, Angelucci V. Load shifting and demand-side management in renewable energy communities: simulations of different technological configurations. Energies 2025;18:872. https://doi.org/10.3390/en18040872
[15] Johannsen RM, Sorknæs P, Sperling K, Østergaard PA. Energy communities' flexibility in different tax and tariff structures. Energy Convers Manag 2023;288:117112. https://doi.org/10.1016/j.enconman.2023.117112
[16] García-Muñoz F, Teng F, Junyent-Ferré A, Díaz-González F, Corchero C. Stochastic energy community trading model for day-ahead and intraday coordination when offering DER's reactive power as ancillary services. Sustain Energy Grids Netw 2022;32:100951. https://doi.org/10.1016/j.segan.2022.100951. https://www.sciencedirect.com/science/article/pii/S2352467722001965.
[17] Ferrara M, Mottola F, Proto D, Ricca A, Valenti M. Local energy community to support hydrogen production and network flexibility. Energies 2024;17:3663. https://doi.org/10.3390/en17153663. https://www.mdpi.com/1996-1073/17/15/3663.
[18] Rebenaque O, Schmitt C, Schumann K, Dronne T, Roques F. Success of local flexibility market implementation: a review of current projects. Util Policy 2023;80:101491. https://doi.org/10.1016/j.jup.2023.101491
[19] Cruz-De-Jesús E, Marano-Marcolini A, Martínez-Ramos J. Participation of energy communities in electricity markets and ancillary services: an overview of successful strategies. Energies 2024;17:4631. https://doi.org/10.3390/en17184631
[20] Rodrigues L, Silva R, Macedo P, Faria S, Cruz F, Paulos J, Mello J, Soares T, Villar J. Planning energy communities with flexibility provision and energy and cross-sector flexible assets. In: 2025 21st international conference on the European energy market (EEM); 2025. p. 1–6. https://doi.org/10.1109/EEM64765.2025.11050164
[21] Imset OH, Ishmam R, Kleinau J, Lavrutich M, Yen D-B. Financial viability of energy communities: the role of flexible assets, balancing market participation and peer-to-peer trading. Energy Econ 2025;152:108985. https://doi.org/10.1016/j.eneco.2025.108985. https://www.sciencedirect.com/science/article/pii/S0140988325008151.
[22] den Berghe LHGJV, Wieczorek AJ. Community participation in electricity markets: the impact of market organisation. Environ Innov Soc Transit 2022;45:302–17. https://doi.org/10.1016/j.eist.2022.10.008
[23] Rodrigues L, Coelho F, Mello J, Villar J. Digital platforms to support the flexibility value chain, run flexibility markets, and Manage energy communities. Curr Sustain Renew Energy Rep 2025;12:15. https://doi.org/10.1007/s40518-025-00264-x
[24] Askeland M, Bjarghov S, Rana R, Morch A, Taxt H. Smart flexibility in energy communities: scenario-based analysis of distribution grid implications and economic impacts. Smart Energy 2025;18:100184. https://doi.org/10.1016/j.segy.2025.100184. https://www.sciencedirect.com/science/article/pii/S2666955225000127.
[25] Siano P, Somma MD, Dolatabadi M. Decentralized optimization for coordinated participation of local energy communities in ancillary services markets with desired energy sharing and costs. Int J Electr Power Energy Syst 2025;168:110694. https://doi.org/10.1016/j.ijepes.2025.110694
[26] ARERA. Prima apertura del mercato per il servizio DI dispacciamento (MSD) alla domanda elettrica e alle unità DI produzione anche DA fonti rinnovabili non già abilitate nonché AI sistemi DI accumulo. Istituzione DI progetti pilota in Vista della costituzione del testo integrato dispacciamento elettrico (TIDE) coerente CON il balancing code europeo; 2017. https://www.arera.it/it/docs/17/300-17.htm.
[27] RSE - Ricerca sul Sistema Energetico. The Italian ancillary services market: MSD and MB mechanisms; 2024 https://www.rse-web.it/wp-content/uploads/2024/03/05_MSD-MB-inglese.pdf [Accessed: 25 June 2025].
[28] Del Greco L, Losi A, Mauro M. Demand response in the Italian regulation and first results. Energia Elettrica Supplement Journal 2022;99. https://doi.org/10.36156/ENERGIA07. https://energiaelettrica.aeit.it/eesj/archi/doc/issue_62.pdf.
[29] Gulotta F, Rossi A, Bovera F, Falabretti D, Galliani A, Merlo M, Rancilio G. Opening of the Italian ancillary service market to distributed energy resources: preliminary results of the UVAM project. In: 2020 IEEE 17th international conference on smart communities: improving quality of life using ICT, IoT and AI (HONET). IEEE; 2020.
[30] Italian Republic. Legislative decree No. 199 of 8 November 2021 — implementation of directive (EU) 2018/2001 on the promotion of the use of energy from renewable sources. 2021. Official Gazette of the Italian Republic, No. 285, 30.11.2021. Accessed:, https://www.gazzettaufficiale.it/eli/id/2021/11/30/21G00194/sg [11 June 2025].